

Q1. Define and classify dental caries. Describe clinical and histological features of dentinal caries.

Q2. Write in detail about the theories of dental caries..

Q3. Write about Histopathology of Enamel Caries.

Q4. Describe immunology of dental caries. Give a brief account of caries activity tests.

Or. Synder test.

Q5. Role of Bacterial Plaque in Dental Caries

Q6. Sequelae of Dental Caries

Q1. Define and classify dental caries. Describe clinical and histological features of dentinal caries.

Dental Caries

Dental caries is an **irreversible progressive microbial disease** of the calcified tissues of the teeth. It is characterized by the **demineralization** of the inorganic portion and the distortion of the organic substances, often leading to **cavitation**.

Classification of Caries

Dental caries can be classified in various ways based on morphology, dynamics, and chronology.

- **Morphology (Anatomical site):**
 - Pit and fissure caries
 - Smooth surface caries
- **Dynamics (Rate of progression):**
 - Acute caries
 - Chronic caries
- **Nature of Attack:**
 - Primary (virgin) caries
 - Secondary (recurrent) caries
- **Chronology:**
 - Infancy (nursing bottle) caries
 - Adolescent caries

Clinical Features of Dentinal Caries

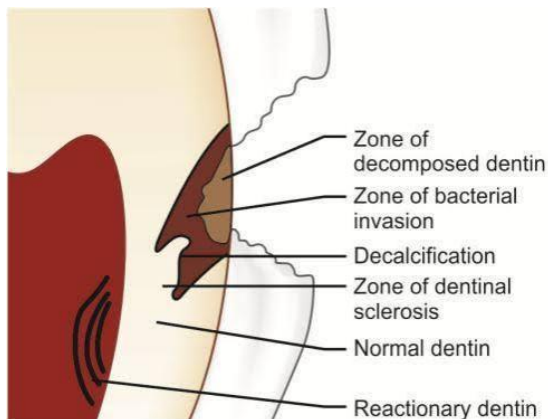
- It occurs on **pit and fissures**, smooth surfaces, and **roots of the teeth**.
- Typically starts in **pit and fissures** with **steep walls** and narrow bases.

- Appears **brown or black**, slightly soft, and catches a fine explorer point.
- The **enamel** bordering the pit and fissure may appear **opaque** and **bluish-white**, often undermined.
- **Lateral spread** of caries at the **DEJ** and into dentin along dentinal tubules can be extensive without fracturing the overhanging enamel.

Histopathological Features of Dentinal Caries

- **Destruction of dentin** occurs by **decalcification** and proteolysis of the **collagenous matrix**.
- Decalcification alone does not cause disintegration but softens the dentin as protein remains intact until destroyed by bacterial enzymes.
- Once the caries reaches the dentin, the lesion spreads rapidly along the **dentinoenamel junctions** due to:
 - Dentin being less mineralized than enamel, making it more vulnerable to acid destruction.
 - Dentin being **cellular**, allowing easier bacterial movement into deeper portions.

Zones of Dentinal Caries



As the lesion progresses, five zones of carious dentin may be distinguished, usually in a **triangular shape** with the apex towards the pulp and the base towards the enamel.

- **Zone 1: Fatty degeneration of Tome's fibers**
 - The innermost layer of the carious dentin with **normal dentinal tubules**.
 - Evidence of **fatty degeneration** of Tome's fibers.
 - No bacteria or crystals in the tubules.
 - **Intertubular dentin** shows normal **collagen** and dense **apatite crystals**.
- **Zone 2: Dentinal Sclerosis**
 - **Deposition of calcium salts** within the dentinal tubules.
 - The superficial layer shows **demineralization** and damage to odontoblastic processes.
 - No bacteria in the tubules.

- Dentin is capable of **remineralization**.
- **Zone 3: Decalcification of Dentin**
 - Narrow zone preceding bacterial invasion, appears **transparent** due to decalcification.
 - Softer than normal dentin.
 - No bacteria in tubules, and **intertubular collagen** remains intact.
 - Capable of **self-repair** and **remineralization**.
- **Zone 4: Bacterial Invasion of Decalcified Dentin**
 - Widened and distorted dentinal tubules filled with **microorganisms**.
 - Very little minerals in the dentin.
 - Denaturation of **collagen fibers**.
 - Cannot undergo **self-repair** or **remineralization**.
 - Must be removed before restoration.
- **Zone 5: Decomposed Dentin**
 - Outermost zone with complete destruction of dentinal tubules.
 - **Liquefaction foci of Miller** occur along dentinal tubules.
 - **Transverse clefts** occur perpendicular to dentinal tubules.
 - Complete destruction of dentin structure leads to **cavitation** at the dentinoenamel junction.

Q2. Write in detail about the theories of dental caries..

The **aetiology** of dental caries is complex and is explained by the following theories:

- **Acidogenic theory**
- **Proteolytic theory**
- **Proteolytic chelation theory**
- **Sucrose chelation theory**
- **Autoimmune theory**

I. ACIDOGENIC THEORY

This theory is also known as **Miller's chemicoparasitic theory**.

It states that dental caries is a **chemicoparasitic process** with two stages:

- **Preliminary stage: Fermentation** of carbohydrates by plaque bacteria produces **organic acids** that **decalcify** enamel and dentin.
- **Later stage:** The **softened residue** dissolves, further progressing the caries. The process is driven by the fermentation of **dietary carbohydrates** like **starches** and **sugars**.
Miller identified four factors influencing caries:
 - **Dietary carbohydrates**
 - **Microorganisms**
 - **Acids**

- **Dental plaque**

Role of Carbohydrates:

- **Fermentable carbohydrates** like glucose, fructose, and **sucrose** are crucial for caries.
- **Sucrose** is the most potent carbohydrate.
- These sugars are fermented by **cariogenic bacteria** to produce acid, leading to **dissolution** of the enamel and dentin.
- The risk of caries increases if the sugar is **sticky, solid, or soft**, as it adheres to the teeth for a long time.
- After sugar ingestion, the **pH of plaque** drops to 4.5 to 5, and neutralization occurs after 10 to 30 minutes.

Role of Microorganisms:

- **Streptococcus mutans** is the most important bacteria involved in caries formation.
- It ferments dietary carbohydrates to produce acid and synthesizes **dextran** from sucrose, aiding the attachment of plaque and acid to the tooth surface.
- **Actinomyces** and **Lactobacillus acidophilus** are also involved, with the latter playing a role in caries progression.

Role of Acids:

- Bacterial fermentation produces acids like **lactic acid, acetic acid, butyric acid, and glutamic acid**, which demineralize enamel and dentin.

Role of Bacterial Plaque:

- **Dental plaque** is a thin film on the tooth surface containing **microorganisms, salivary mucin, food debris, and leucocytes**.
- Plaque initiates caries by:
 - Harboring **cariogenic bacteria**.
 - Retaining acids on the tooth surface.
 - Protecting acids from neutralization by saliva.

II. PROTEOLYTIC THEORY

This theory suggests that **proteolytic enzymes** released by cariogenic bacteria break down the **organic matrix** of enamel, causing the inorganic crystals to detach, leading to cavity formation.

- **Gottlieb and Gottlieb** and others stated that **cariogenic bacteria** destroy the organic components in enamel through **proteolysis**, accompanied by acid formation.

- However, this theory doesn't explain the role of **sucrose**, **pH**, and **fluoride** in caries.
- It suggests that **proteolysis** is not significant in the initiation of caries but may contribute to **progression**.

III. PROTEOLYTIC CHELATION THEORY

Proposed by **Schatz et al (1955)**, this theory suggests that both **proteolytic breakdown** of the enamel matrix and **chelation** of minerals occur during caries.

- Proteolytic breakdown of the **organic portion** of enamel occurs first.
- This is followed by the formation of a **chelating agent** (e.g., food debris, bacterial products) that releases **calcium ions** from enamel and dentin, leading to tooth decay.
- Both organic and inorganic portions of enamel are attacked simultaneously.

IV. SUCROSE CHELATION THEORY

Proposed by **Egglers-Lura (1967)**, this theory states that **sucrose** itself, not the acid from it, causes enamel dissolution by forming **ionized calcium saccharate**.

- High concentrations of sucrose lead to the formation of **complex substances** like **calcium saccharide**, which release calcium and phosphorus ions from enamel, causing decay.

V. AUTOIMMUNE THEORY

This theory suggests that **autoimmune mechanisms** may damage specific **odontoblast cells** within the pulp of certain teeth, leading to caries.

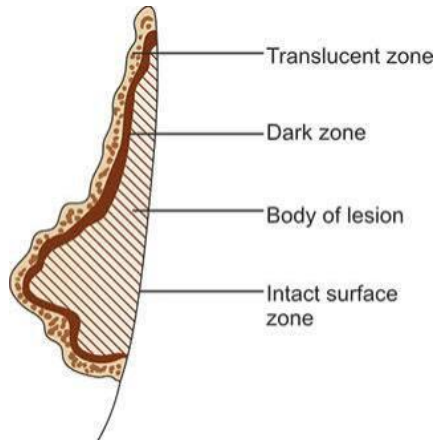
Q3. Write about Histopathology of Enamel Caries.

Early Enamel Caries

- **Loss of interprismatic/inter-rod substances**, making the **enamel rods more prominent**.
- A **dark line** often appears at **right angles** to enamel rods, indicating **segments**.
- **Accentuation of incremental striae of Retzius** is commonly observed.
- **Accentuation of perikymata** and striae of Retzius are **early microscopic changes**.

Advanced Enamel Caries

- There are four **zones** visible in advanced enamel caries:



1. **Translucent zone**
2. **Dark zone**
3. **Body of the lesion**
4. **Surface zone**

Zone 1: Translucent Zone

- This is the **deepest zone**, located at the advancing front of the enamel lesion.
- It is the **first recognizable change** from normal enamel.
- This zone is **more porous** than normal enamel, with pores located at **prism boundaries** and other junctional sites.
- The **pore volume** is about **1%**, and it appears **structureless**.
- It contains more **fluoride** than normal enamel.

Zone 2: Dark Zone

- Also called the **positive zone**.
- Located just above the **translucent zone**.
- Appears **dark** due to excessive **demineralization** of the enamel.
- The zone is **narrower** in rapidly advancing caries and **wider** in slower lesions.
- Contains **2-4% pore volume**, with pores larger than in normal enamel but smaller than in the translucent zone.
- Shows some **remineralization** of the carious lesion.

Zone 3: Body of the Lesion

- Situated between the **dark zone** and the **surface layer** of enamel.
- Represents the area of **greatest demineralization**.
- The **pore volume** is between **5-25%**.

Zone 4: Surface Zone

- The surface zone, when viewed under **polarizing light**, appears relatively **unaffected**.
- This may be due to surface **remineralization** by **salivary mineral ions**.
- The **pore volume** is less than **5%**.
- The surface zone is usually **demineralized** when the lesion reaches the **dentin**.

Q4. Describe immunology of dental caries. Give a brief account of caries activity tests.

Or. Synder test.

Immunology of Dental Caries

- The theory summarizing the role of **bacteria, sugars, and teeth** in dental caries is as follows:
Bacteria + Sugars + Teeth → Organic acids → Dental Caries.

Caries Activity Tests

- Several caries activity tests have been developed to assess the risk of dental caries.

I. Lactobacillus Colony Test

- The **most widely used microbiological test** for caries activity.
- Also known as **LB colony count**, it measures the number of **aciduric bacteria** in the saliva by counting colonies on **Rogosa medium**.

II. Snyder Test

- **Purpose:** Measures the ability of **salivary microorganisms** to produce **organic acids** from the **metabolism** of **carbohydrates**.

Materials Used

- **Glucose agar media** with **bromocresol green indicator dye**.
- **Indicator dye** changes color depending on the **pH** of the medium.

Principle

- The dye changes from green to yellow within the **pH range of 5.4 to 3.8**.
- This indicates the production of **acidic by-products** by microorganisms in the saliva, which correlates to **caries activity**.

Procedure

- **Paraffin-stimulated saliva** is collected from the patient.
- The saliva is mixed into the **Snyder test agar medium**.
- The medium initially appears **blue-green** (pH 4.7–5).
- If the **microorganisms are active**, they metabolize the **glucose**, producing acids.
- The **color changes** from **blue-green to yellow** (pH 4), indicating **high caries activity**.

Significance

- A faster color change to yellow shows **higher acid production** and indicates **increased risk of dental caries**.

III. Swab Test

- Similar to the Snyder test, but uses an **oral swab** to sample the **oral flora**.
- The swab is incubated, and a **color comparator** is used to measure pH changes.
- A pH of **4.1 and below** indicates **marked caries activity**, while **4.5 to 4.6** suggests **slightly active** caries.

IV. Salivary S. mutans Level Test

- Involves incubation of plaque samples on **Mitis Salivarius Agar (MSA)**.
- The agar plates are incubated at **37°C** for **48 hours**.
- **Levels of S. mutans** higher than **10⁵** indicate an unacceptable cariogenic challenge.

V. S. mutans Dip Slide Method

- Salivary samples are classified based on the density of **S. mutans colonies** on modified **Mitis Salivarius Agar**.
- Saliva is collected after **5 minutes** of stimulation and poured over the agar.
- **S. mutans** colonies are visible as **blue** colonies, and the colony density is compared with a standard chart to classify the level.
 - **0**: Negligible
 - **1**: <100,000
 - **2**: 100,000 - 1,000,000
 - **3**: >1,000,000

VI. Buffer Capacity Test

- Measures the amount of acid required to lower the **pH** of saliva.

- A **pH meter** or **color indicators** is used to evaluate the pH.
- A **lower buffering capacity** of saliva indicates a higher **caries risk**.

VII. Enamel Solubility Test

- Measures the amount of **enamel dissolved** in 4 hours when the patient's saliva is mixed with **glucose** and **powdered enamel**.
- The enamel is dissolved by the **acid** formed.
- This test is not suitable for **office procedures**.

VIII. Salivary Reductase Test

- Measures the activity of the **reductase enzyme** present in salivary bacteria.
- **Paraffin-stimulated saliva** is collected, and **Diazoresorcinol** is added as an indicator dye, turning the saliva **blue**.
- The **reductase enzyme** causes color changes from blue to other colors, indicating the **cariogenic potential** of the patient's saliva.

Q5. Role of Bacterial Plaque in Dental Caries

- **Dental plaque harbors caries-causing microorganisms** and keeps them in contact with tooth surfaces.
- **Plaque contains carbohydrates and lipids** that help bacteria adhere and serve as a source of fermentable substrates.
- Acts as a **barrier to salivary buffers**, preventing acids in plaque from being neutralized.
- **Certain cariogenic bacteria store polysaccharides**, which act as a carbohydrate reserve during sugar-free diets.
- Plaque contributes to the **initiation of dental caries** through:
 - Harboring **cariogenic bacteria** on tooth surfaces.
 - Retaining **acids on tooth surfaces** for a long time.
 - Protecting acids from **neutralization by saliva's buffering action**.
- The **acquired pellicle**, a glycoprotein from saliva, forms an important part of plaque and serves as a nutrient for microorganisms.

Q6. Sequelae of Dental Caries

Dental caries, if left untreated, leads to **progressive destruction** of tooth structure and can cause various complications. These are the sequelae:

1. Pulpal Involvement

- **Reversible Pulpitis:** Mild inflammation; can heal if treated early.
- **Irreversible Pulpitis:** Severe pain due to persistent inflammation.
- **Pulp Necrosis:** Death of the pulp tissue, leading to loss of vitality.

2. Periapical Pathologies

- **Acute Periapical Abscess:** Pus formation at the tooth apex causing pain and swelling.
- **Chronic Periapical Granuloma:** Localized inflammation with granulation tissue formation.
- **Radicular Cyst:** Fluid-filled cavity at the root apex, lined by epithelium.

3. Spread of Infection

- **Osteomyelitis:** Infection spreads to the jaw bone, causing severe pain and bone destruction.
- **Cellulitis:** Diffuse infection of the soft tissues, resulting in swelling and redness.
- **Ludwig's Angina:** Life-threatening condition with bilateral cellulitis in the submandibular region.

4. Tooth Loss

- Extensive destruction of the tooth structure leads to functional and aesthetic issues.

5. Systemic Complications

- **Bacteremia:** Bacteria entering the bloodstream.
- **Endocarditis:** Infections affecting the heart valves in susceptible individuals.

Key Points

- Early detection and treatment of dental caries prevent these complications.
- Pulpal and periapical conditions often require endodontic treatment or extraction.